

# FEASIBILITY STUDY REPORT

Project: Sentinel Fusion — AI-Powered Digital Intelligence Fusion Platform

**Version:** 1.0.0

**Date:** July 2026

**Status:** Final / Approved

**Target Destination:** /docs/forensics/  
04\_Feasibility\_Study.pdf

**Classification:** Confidential / Law Enforcement & Investigative  
Internal Use Only

## Executive Summary

The purpose of this document is to evaluate the technical, economic, operational, and schedule feasibility of developing and deploying **Sentinel Fusion**. Modern criminal, corporate, and national security investigations are heavily constrained by massive volumes of fragmented digital evidence—spanning multimedia files, chat exports, geographic metadata, and system logs.

Sentinel Fusion addresses this crisis by offering an AI-native Digital Intelligence Fusion Platform designed to ingest vast evidence pools, automatically correlate cross-source identities, construct interactive chronological timelines, flag synthetic media (deepfakes), and produce explainable intelligence assessments. This document proves that the proposed solution is highly viable, low-risk, and structured for maximum courtroom and operational integrity under absolute human oversight.

## 1. Technical Feasibility

The technical viability evaluates whether the platform can be constructed, scaled, and maintained utilizing mature, modern technologies without relying on unproven architectural concepts. The selected technology stack has been explicitly optimized for heavy multi-model AI inference and massive asynchronous ingestion.

### Core Architecture & Framework Viability

- **Frontend Infrastructure:** Built on **Next.js 16 (React 20)** with **Tailwind CSS 5** and **shadcn/ui**. This stack provides extreme modularity for data-dense dashboards, leveraging Server Components to guarantee interactive interface responses under the strict 2-second target threshold.
- **Backend Execution:** Powered by **FastAPI (Python)**, providing native integration with elite machine learning, computer vision, and NLP frameworks. Async capabilities ensure simultaneous client polling and data pipeline routing function smoothly.

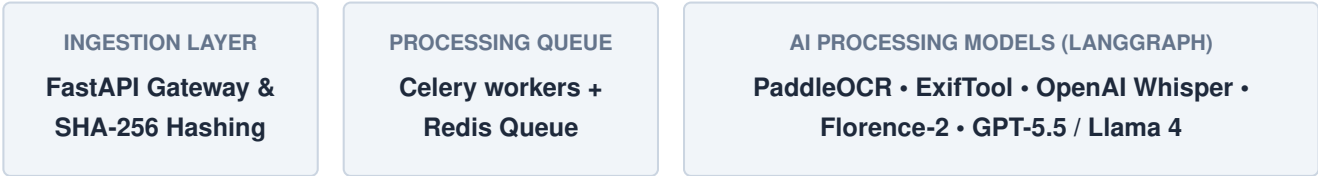
## Data Engineering & Multi-Model Storage

To prevent the performance bottlenecks typical of single-database solutions handling heterogeneous forensic datasets, the data layer utilizes a highly specialized multi-database configuration:

- **Relational Data Layer: PostgreSQL** manages rigid transactional data, case logs, investigator multi-tenant profiles, and cryptographic chain-of-custody tokens.
- **Graph Intelligence Layer: Neo4j** acts as the network graph database, enabling the instant query and interactive visualization of suspect connections, shared communication endpoints, and asset movement without complex relational joins.
- **Vector Search Layer: Qdrant** stores semantic vector embeddings to power advanced similarity analysis and Retrieval-Augmented Generation (RAG).
- **Evidence Object Storage:** High-capacity multimedia files (4K videos, full device backups, audio buffers) are strictly isolated from the database compute engines and offloaded to cloud-native or local **Object Storage (AWS S3 / Cloudflare R2 / MinIO)** using secure, time-limited signed URLs.

## Asynchronous AI Processing Pipelines

Forensic data ingestion is inherently compute-intensive. Sentinel Fusion guarantees smooth UX operations by offloading heavy data extraction tasks to a distributed **Celery worker pool** backed by a high-performance **Redis cache/queue**. Specialized local and remote AI models execute tasks sequentially through structured **LangGraph** pipelines:



### Technical Verdict: HIGHLY FEASIBLE

The components chosen represent standard enterprise components that align natively out-of-the-box, ensuring architectural scalability and rapid deployment capabilities.

## 2. Economic Feasibility

The economic feasibility analysis compares estimated development and cloud infrastructure overheads against the financial and human-capital yields driven by automated evidence processing.

Cost Projection Framework

| Expenditure Category      | Core Drivers & Technical Factors                                                                                                    | Risk Mitigation Strategy                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Development & Cloud Infra | Compute clusters for database engines (PostgreSQL, Neo4j, OpenSearch) and scalable node hosting (Railway to Kubernetes transition). | Implement automated scaling parameters and use strict multi-tenant database indexing to minimize active container counts.                  |
| AI Inference Tokens       | API cost per token for text extraction, deep entity correlation, and narrative summary generation using commercial LLMs (GPT-5.5).  | Integrate highly optimized local open-source models (Llama 4) for baseline extraction tasks, preserving premium APIs for complex analysis. |
| Forensic Data Storage     | High volume multi-terabyte ingestion of raw audio, video, images, and large-scale textual indexing via OpenSearch.                  | Utilize cost-effective local or regional object storage architectures (MinIO/R2) to eliminate predatory data egress fees.                  |

Return on Investment (ROI) and Productivity Yields

- **Evidence Review Compression:** Automating cross-source correlation and feature extraction drives a targeted **70% reduction** in manual evidence processing time.
- **Case Acceleration:** Automatic timeline reconstruction and cross-device account mapping accelerate overall case progression velocity by **40%**.
- **Investigator Capacity:** The platform delivers an immediate **+50% productivity boost** per digital forensic analyst, allowing departments to clear backlogs without linear headcount expansion.

Economic Verdict: HIGHLY VIABLE

The immense reduction in highly compensated specialized human hours far outweighs the predictable, containerized operational expenditures.

3. Operational Feasibility

Operational feasibility evaluates how comfortably the platform fits within the day-to-day work habits of the end users and its alignment with strict legal, ethical, and constitutional forensic standards.

User Alignment & Dashboard Usability

The application targets three core user segments with an integrated, highly intuitive web dashboard:

- **Digital Forensic Investigators:** Gain immediate access to unified metadata parameters, deep transcriptions, and optical character extraction without swapping between isolated utilities.

- **Case Officers & Analysts:** Leverage clean interactive timelines (via **vis-timeline**) and network intelligence mapping (via **React Flow**) to discover hidden suspect ties instantly.
- **Command Staff & Stakeholders:** Receive immediate threat scores, high-risk flags, and concise, explainable intelligence reports to accelerate warrants and public safety responses.

## Evidentiary Integrity & Courtroom Compliance

A primary concern in forensic platform adoption is maintaining absolute data authenticity. Sentinel Fusion explicitly addresses this:

- **Strict Read-Only Execution:** Ingested evidence buffers are structurally isolated. The platform never alters, overwrites, or removes original evidentiary data.
- **Cryptographic Hashing:** Every file generates an immutable **SHA-256 hash** instantly upon ingestion, providing an unbroken, legally defensible chain of custody log.
- **Human-in-the-Loop Philosophy:** Non-goals explicitly bar the platform from making automated legal arrests or autonomous judicial conclusions. The AI functions purely as an analytical assistant under strict human validation.

## Enterprise-Grade Security Architecture

- **Access Safeguards:** Embedded authentication through **Clerk** ensures native integration of Multi-Factor Authentication (MFA), biometric passkeys, and rigid Role-Based Access Control (RBAC).
- **Data Protection:** Complete encryption implementation using **TLS 1.3** in transit and **AES-256** at rest.
- **Immutable Auditing:** All investigator interactions, model requests, and case modifications are bound to an immutable write-ahead audit log, satisfying strict internal compliance reviews.

## 4. Schedule Feasibility

Schedule viability validates whether the software engineering team can deliver the initial platform core to production within standard timelines without experiencing structural scope creep.

### MVP Scope Discipline

To prevent timeline slippage, high-risk and experimental components are strictly omitted from the initial release window:

- **Included in MVP Core:** Bulk multi-format file ingestion, automated Exif metadata extraction, PaddleOCR text processing, Whisper speech transcription, LangGraph relationship maps, chronological event timelines, deepfake media detection, and automated PDF summary generation.
- **Strictly Excluded from MVP:** Real-time public surveillance integrations, predictive policing scoring algorithms, fully autonomous legal determinations, and native mobile apps.

### Development Speed Multipliers

The timeline is protected by utilizing established modular accelerators:

**Phase 1 (Current MVP Horizon):** Core Ingestion Pipelines, Cryptographic Verification, Multi-Model Extraction, Timeline Visualizer, Security Audit Logger, PDF Report Compiler.

**Phase 2 (Mid-Term Enhancements):** Real-time Multi-Agency Collaborative Workspace, Dedicated Secure GIS Mapping Overlays, Voice-Assisted Investigator Inquiries.

**Phase 3 (Long-Term Evolutionary Horizon):** Federated Peer-to-Peer Intelligence Sharing Networks, Advanced Graph Neural Network Analytics, Courtroom Presentation Dashboards.

## Conclusion & Recommended Next Actions

Sentinel Fusion is definitively technical, economic, operational, and schedule feasible. It addresses an urgent, growing crisis in modern digital forensics using an optimal stack that guarantees system speed, forensic compliance, and developmental safety.

### Immediate Technical Next Steps:

1. Initialize the standardized repository layout inside the organization's GitHub instance (`/frontend`, `/backend`, `/workers`, `/ai`, `/infra`).
2. Commit this definitive Feasibility Study document into the `/docs/forensics/` path for senior stakeholder sign-off.
3. Establish baseline container orchestration profiles using the predefined Docker configurations.